PRODUCT MANUAL

NEXORA SENTINEL

User Guide & Product Manual

ESP32-CAM Architecture · Event-Driven Security





Welcome to Nexora

Intelligent protection for your personal space.

Nexora is designed to bring intelligent monitoring to your wardrobe through door-event detection, automatic video recording, local storage and connected notifications.

Built around an ESP32-CAM-based system, Nexora combines embedded computing, camera technology, magnetic door sensing and wireless connectivity in a compact security solution.

The system is designed to capture activity associated with wardrobe access, preserve video recordings locally and provide notifications through Telegram when the connected features are configured and available.

Your Nexora system

- Door-event detection: Detects changes in the wardrobe door's position through a magnetic sensor.
- Automatic recording: Initiates recording in response to a detected opening event.
- Local video storage: Stores recorded footage on a microSD card.
- Connected notifications: Supports Telegram notifications and video delivery.
- Compact integration: Combines sensing, processing, recording and connectivity in one system.

Designed around three principles

Sense. Detect relevant changes in the environment.

Understand. Process the detected event and initiate the appropriate response.

Act. Record the event and communicate information through the configured notification system.

Thank you for choosing Nexora.

Meet Your Nexora System



System components

Nexora integrates several electronic and software components to deliver its intended functionality.

1. Camera and processing unit

The ESP32-CAM-based controller provides the computing and camera interface used by the system. It processes camera data and coordinates recording and communication tasks.

2. Magnetic door sensor

A magnetic reed switch detects the relative position of the wardrobe door and its corresponding magnet. The sensor provides the door-state input used to initiate the recording workflow.

3. Camera module

The camera captures image frames for video recording. The prototype configuration uses VGA resolution of 640 × 480 pixels at 5 frames per second.

4. microSD storage

A microSD card provides local storage for recorded video files. Recordings can be retained locally even when a separate video-delivery attempt does not complete, subject to the final firmware's storage and error-handling behavior.

5. Wi-Fi connectivity

Wi-Fi enables the system to communicate with the configured network and access connected services.

6. Telegram integration

Telegram integration is intended to deliver event notifications and recorded video to a designated chat, subject to correct configuration, network availability and successful transmission.

How the components work together

The magnetic sensor detects a door event. The controller processes the event, coordinates camera recording, saves the resulting footage and initiates the configured notification workflow.

Each component contributes to the overall operation of the system.

Installation and Placement

Positioning your Nexora system

Correct placement helps ensure that the camera has a clear view of the intended area and that the door sensor can detect changes in door position.

Step 1: Choose the installation position

Select a stable position within or near the wardrobe that provides the intended camera view.

Ensure that:

- The camera lens is not obstructed by clothing, shelves or the wardrobe frame.
- The controller and its connections are accessible for maintenance.
- The installation does not interfere with normal wardrobe use.
- Cables are kept clear of moving doors and hinges.

Step 2: Position the magnetic sensor

The sensor and magnet should be mounted on opposing surfaces that move relative to one another when the door opens.

When the door is closed, the two components should align according to the final installation design. When the door opens, their relative position changes, allowing the system to detect the event.

Step 3: Secure the components

Use the mounting method supplied with the final product. Ensure that the sensor, magnet, camera and cables remain stable during normal door movement.

Step 4: Check the installation

Before relying on the system, verify that:

- The door opens and closes without pulling or pinching cables.
- The sensor and magnet remain correctly aligned when the door is closed.
- The camera view remains unobstructed.
- The installation remains stable throughout the door's movement.

Power connection

Connect the system only using the power supply specified for the final Nexora product. The approved power input, connector type and adapter specification should match the product documentation supplied with the unit.

First-Time Setup

Preparing your Nexora system

Before first use, confirm that the system's physical components are installed and that the required storage and connectivity are available.

1. Inspect the system
Check the camera assembly, sensor, mounting components, cable and power connector for visible damage.

2. Prepare local storage
Insert a compatible microSD card according to the final product's supported card capacity, filesystem and formatting requirements.
The exact supported card specifications should be confirmed in the product documentation for the supplied unit.

3. Connect power
Connect the approved power supply and allow the system to complete its startup process.

4. Configure connectivity
Connect the system to the supported Wi-Fi network using the setup procedure provided for the final product.
Network credentials should be kept private and shared only with trusted administrators.

5. Configure Telegram
Complete the approved Telegram setup process for the intended recipient chat.
Confirm that the correct chat is configured before relying on event notifications or video delivery.

6. Perform a functional test

Test the complete intended workflow:

1. Close the wardrobe door.
2. Open the door and verify that the event is detected.
3. Confirm that recording begins.
4. Close the door and allow the recording workflow to finish.
5. Check whether the video is saved to the microSD card.
6. Verify whether the expected Telegram notification and video arrive.

TIP: A successful setup should be confirmed through actual testing rather than inferred from the device appearing online.

Everyday Operation

Understanding the recording workflow

Nexora is designed around door-triggered recording.

Door closed

When the wardrobe door is closed, the magnetic sensor provides the corresponding door-state input. The system remains available to detect a subsequent opening event.

Door opened

When the door opens and the sensor detects the change, the system is intended to initiate the recording workflow.

The camera captures frames, and the controller processes the video data for storage.

Door closed again

When the door closes, the system is intended to conclude the event recording and complete the associated video-storage workflow.

Recording and notification

Following the recording event, the system is intended to save the video locally and initiate the configured Telegram notification and video-delivery process.

Successful delivery depends on the network connection, Telegram availability, file size and completion of the upload process.

Understanding the two delivery paths

Local recording: Video is stored on the microSD card according to the system's recording and storage behavior.

Connected delivery: Telegram is used for remote notifications and video delivery when the connection and configuration permit it.

These functions serve different purposes. Local storage provides an on-device copy, while connected delivery provides remote access to event information.

The exact recording limits, retry behavior and retention policy depend on the final product configuration.

Video Storage and File Management

Managing your recordings

The microSD card is the system's local video-storage medium.

Recordings should be handled carefully to preserve the original files and maintain the availability of important footage.

Accessing recorded video

Follow the approved shutdown procedure before removing the microSD card.

Once removed, the card can be accessed using a compatible card reader. Copy required files to a secure computer or storage location and verify that the copied files can be opened.

Preserving recordings

To reduce the risk of losing footage:

- Back up important recordings before performing storage maintenance.
- Keep original files until the backup has been verified.
- Avoid formatting the card while required recordings remain on it.
- Store copied recordings in a secure location.
- Limit access to recordings to authorized users.

Storage capacity

Available recording space depends on the microSD card's capacity, the amount of recorded footage and the size of individual video files.

The final product documentation should specify the supported card capacity, format, storage-full behavior and recommended card-management procedure.

Video delivery and local storage

A Telegram delivery failure does not necessarily mean that a locally saved recording has been lost. Check the microSD card according to the approved retrieval procedure before taking any action that could affect stored footage.

Telegram Notifications and Connectivity

Connected event notifications

Nexora is designed to integrate with Telegram to communicate information about detected wardrobe events.

When the connected workflow is configured and functioning, it is intended to deliver an event notification and associated video to the designated Telegram chat.

Requirements for connected operation

Successful delivery depends on:

- Correct Wi-Fi configuration.
- A functioning network connection.
- Correct Telegram setup and recipient selection.
- Availability of the Telegram service.
- Successful completion of the video-transfer process.

If a notification does not arrive

Check the network connection and confirm that the intended Telegram chat is correctly configured.

Then inspect local storage to determine whether a recording was created.

If a notification arrives without a video

Check whether the corresponding video exists on the microSD card. Video delivery and local recording should be assessed separately.

If a video transfer fails

Preserve the local recording and follow the final product's approved troubleshooting procedure. Avoid deleting or formatting files before confirming that a usable copy exists.

Protecting account credentials

Keep Wi-Fi passwords, Telegram bot tokens and other configuration credentials private. Do not include these credentials in photographs, public documentation or messages shared with untrusted recipients.

Troubleshooting

Common issues and recommended checks

Issue	Recommended Check & Action
The system does not detect a door opening	Check that the sensor and magnet are correctly positioned and remain aligned when the door is closed. Verify the installation and sensor connections according to the approved product procedure.
No recording is available	Check the microSD card and confirm that the card is recognized and has available storage. Verify the complete door-event workflow and inspect whether the event was detected.
A recording exists but cannot be opened	Copy the original file before attempting any recovery operation. Test the copied file using a compatible media player. Do not format the card if the original recording is still required.
Telegram notifications are missing	Confirm the network connection, Telegram configuration and intended recipient chat. Check local storage independently to establish whether the event was recorded.
Video delivery is incomplete	Check the available local copy before attempting further troubleshooting. Follow the final product's approved procedure for handling unsuccessful transfers.
The system does not reconnect	Check the availability of the configured Wi-Fi network and follow the supported network setup procedure. Avoid repeated power cycling without first identifying the cause of the connection problem.
The system behaves inconsistently	Record the circumstances of the issue, including the approximate time, door state, recording behavior and whether a local file exists. This information can help identify whether the issue relates to sensing, recording, storage or network delivery.

Maintenance, Privacy and Responsible Use

Routine maintenance

Periodic inspection helps maintain the physical installation and identify problems before they affect normal use.

Camera and enclosure
Keep the lens clean using a soft, dry microfiber cloth. Inspect the enclosure and mounting points for looseness or damage.

Sensor and magnet
Check that the sensor and magnet remain aligned when the wardrobe is closed and separate appropriately when it opens.

Cables and connections
Inspect accessible cables and connectors for visible damage, looseness or interference with moving parts.

Storage
Back up important recordings and follow the approved microSD maintenance procedure.

Operating environment
Keep the system within the environmental conditions specified for the final product. Installation inside an enclosed wardrobe should follow the validated mounting, ventilation and operating requirements.

Privacy and responsible use

Nexora is intended to monitor a designated area associated with wardrobe access.

Users should:

- Inform affected household members where appropriate.
- Comply with applicable privacy and recording laws.
- Restrict access to recordings and connected accounts.
- Avoid recording areas beyond the intended purpose of the system.
- Establish suitable retention and deletion practices for stored footage.

Access and information security

Keep account credentials and configuration details secure. Review who can access the configured Telegram chat and any location where recorded footage is stored.

Product Reference and Support

Technical reference

Feature	Description
Product	Nexora Wardrobe Security Camera
Processing platform	ESP32-CAM-based system
Door sensing	Magnetic reed switch
Prototype sensor input	GPIO 13
Prototype recording profile	VGA, 640 × 480 pixels, 5 FPS
Prototype video handling	AVI container with JPEG-compressed frames
Local storage	microSD card
Connectivity	Wi-Fi
Connected notifications	Telegram integration
Power input	Refer to the approved specification supplied with the product
Supported microSD capacity	Refer to the approved product specification

Frequently asked questions

Does Nexora store video locally?

The system is designed to store recorded video on a microSD card. Confirm the presence and integrity of a recording using the approved retrieval procedure.

Does Telegram delivery guarantee that every video arrives?

No. Connected delivery depends on network availability, service availability and successful completion of the upload process.

Can recorded video be retrieved without Telegram?

A locally saved recording can be accessed through the approved microSD retrieval procedure, subject to the recording being present and readable.

Can the product be installed in any wardrobe?

Installation suitability depends on the mounting arrangement, camera view, power access and the final product's validated environmental and operating requirements.

Product information

Product name	Nexora Sentinel
Model number	2026101701
Manufacturer / product owner	Kanishk Lahoti @Ceo Nexora
Customer support	+91 8369125079 or kanishklahoti9@gmail.com
Warranty	1 years
Firmware Version	V1.5 UI 2
Product documentation version	V3

NEXORA

Sense. Understand. Act.

